

A Literature-Implied Answer to a Parseval-Frame Class Problem

Run fa_banach_001, lane 2

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Status

literature_implied_answer. This packet records that a finite Parseval-frame equivalence-class subproblem stated in arXiv:0906.5606 is answered by the standard projection/Gramian parametrization of normalized tight frames, stated explicitly in Holmes–Paulsen [2].

Source Problem

The source is Casazza–Kutyniok–Li–Rozell [1], arXiv:0906.5606. In Section 4, subsection *Extensions and Related Problems*, under *Equivalence Classes*, the authors write that the following apparently simple problem is still unsolved: construct one Parseval frame in each equivalence class, choosing unitary equivalence as the relation.

Supporting Literature

Holmes and Paulsen [2] identify finite normalized tight frames with rank projections through the analysis operator. In their notation, an (n, k) -frame has analysis isometry $V : \mathbb{F}^k \rightarrow \mathbb{F}^n$, and VV^* is the Gramian/correlation projection. They state that there is a one-to-one correspondence between $n \times n$ rank k projections and type-I equivalence classes of (n, k) -frames. Type-I equivalence is exactly unitary equivalence of the ordered frame vectors.

Identification and Representative

Let P be a rank k orthogonal projection on $\mathbb{F}^n$, where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. Define

$$f_i = Pe_i \in \text{ran } P, \quad i = 1, \dots, n.$$

Then $\{f_i\}_{i=1}^n$ is a Parseval frame for $\text{ran } P$, because for every $x \in \text{ran } P$,

$$\sum_{i=1}^n |\langle x, Pe_i \rangle|^2 = \sum_{i=1}^n |\langle Px, e_i \rangle|^2 = \sum_{i=1}^n |\langle x, e_i \rangle|^2 = \|x\|^2.$$

Moreover, the Gram matrix of this frame is

$$(\langle f_j, f_i \rangle)_{i,j} = (\langle Pe_j, Pe_i \rangle)_{i,j} = P.$$

Conversely, if $\{g_i\}_{i=1}^n$ is a Parseval frame for a k -dimensional Hilbert space with analysis isometry V , then $P = VV^*$ is a rank k projection and $Vg_i = Pe_i$. Thus the original frame is unitarily equivalent to the projection representative $\{Pe_i\}_{i=1}^n$.

Scope

This gives one representative in each finite ordered Parseval-frame unitary-equivalence class. It does not solve the source's broader fusion-frame equivalence-class program, nor its separate weighted, chordal-distance, or sparsity questions.

References

- [1] P. G. Casazza, G. Kutyniok, S. Li, and C. J. Rozell, *Fusion frames: existence and construction*, arXiv:0906.5606.
- [2] R. B. Holmes and V. I. Paulsen, *Optimal frames for erasures*, Linear Algebra Appl. 377 (2004), 31–51. DOI: 10.1016/j.laa.2003.07.012.